

Theorem (Cauchy–Schwarz Inequality). *Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be real numbers. Then*

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

with equality iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i .

Proof. **1. Inner-product notation.** Define the vectors

$$\mathbf{a} = (a_1, \dots, a_n), \quad \mathbf{b} = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

The standard Euclidean inner product and norm are

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=1}^n a_i b_i, \quad \|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a} \rangle = \sum_{i=1}^n a_i^2, \quad \|\mathbf{b}\|^2 = \sum_{i=1}^n b_i^2.$$

2. A non-negative quadratic. For an arbitrary real parameter t consider

$$Q(t) = \|\mathbf{a} - t\mathbf{b}\|^2 = \sum_{i=1}^n (a_i - tb_i)^2.$$

Since a squared norm is non-negative,

$$Q(t) \geq 0 \quad \text{for all } t \in \mathbb{R}. \quad (1)$$

3. Expansion of $Q(t)$. Using $(x - ty)^2 = x^2 - 2txy + t^2y^2$ we obtain

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2ta_i b_i + t^2 b_i^2) \\ &= \|\mathbf{a}\|^2 - 2t\langle \mathbf{a}, \mathbf{b} \rangle + t^2 \|\mathbf{b}\|^2. \end{aligned}$$

Thus $Q(t)$ is the quadratic polynomial

$$Q(t) = \|\mathbf{b}\|^2 t^2 - 2\langle \mathbf{a}, \mathbf{b} \rangle t + \|\mathbf{a}\|^2.$$

4. Discriminant condition. If $\|\mathbf{b}\| \neq 0$ then the leading coefficient $\|\mathbf{b}\|^2 > 0$. A quadratic $At^2 + Bt + C$ with $A > 0$ that is non-negative for every real t must have non-positive discriminant:

$$B^2 - 4AC \leq 0.$$

Applying this to the coefficients above gives

$$(-2\langle \mathbf{a}, \mathbf{b} \rangle)^2 - 4\|\mathbf{b}\|^2 \|\mathbf{a}\|^2 \leq 0,$$

i.e.

$$4\langle \mathbf{a}, \mathbf{b} \rangle^2 \leq 4\|\mathbf{a}\|^2 \|\mathbf{b}\|^2.$$

Cancelling the positive factor 4 yields

$$\langle \mathbf{a}, \mathbf{b} \rangle^2 \leq \|\mathbf{a}\|^2 \|\mathbf{b}\|^2,$$

which is exactly

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right). \quad (2)$$

5. Zero-vector case. If $\|\mathbf{a}\| = 0$ then all $a_i = 0$ and both sides of (2) are 0. Similarly, if $\|\mathbf{b}\| = 0$ the right-hand side is 0 while the left-hand side is also 0. Hence the inequality holds trivially when either vector is the zero vector.

6. Equality condition. Equality in (2) occurs precisely when the discriminant is zero, i.e.

$$(-2\langle \mathbf{a}, \mathbf{b} \rangle)^2 = 4\|\mathbf{a}\|^2 \|\mathbf{b}\|^2.$$

In this case the quadratic $Q(t)$ has a unique root

$$t_0 = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{b}\|^2},$$

and because $Q(t_0) = 0$ we have $\|\mathbf{a} - t_0 \mathbf{b}\|^2 = 0$, whence $\mathbf{a} = t_0 \mathbf{b}$. Thus there exists $\lambda \in \mathbb{R}$ with $a_i = \lambda b_i$ for all i . Conversely, if the vectors are proportional the inequality becomes an equality. $\square$